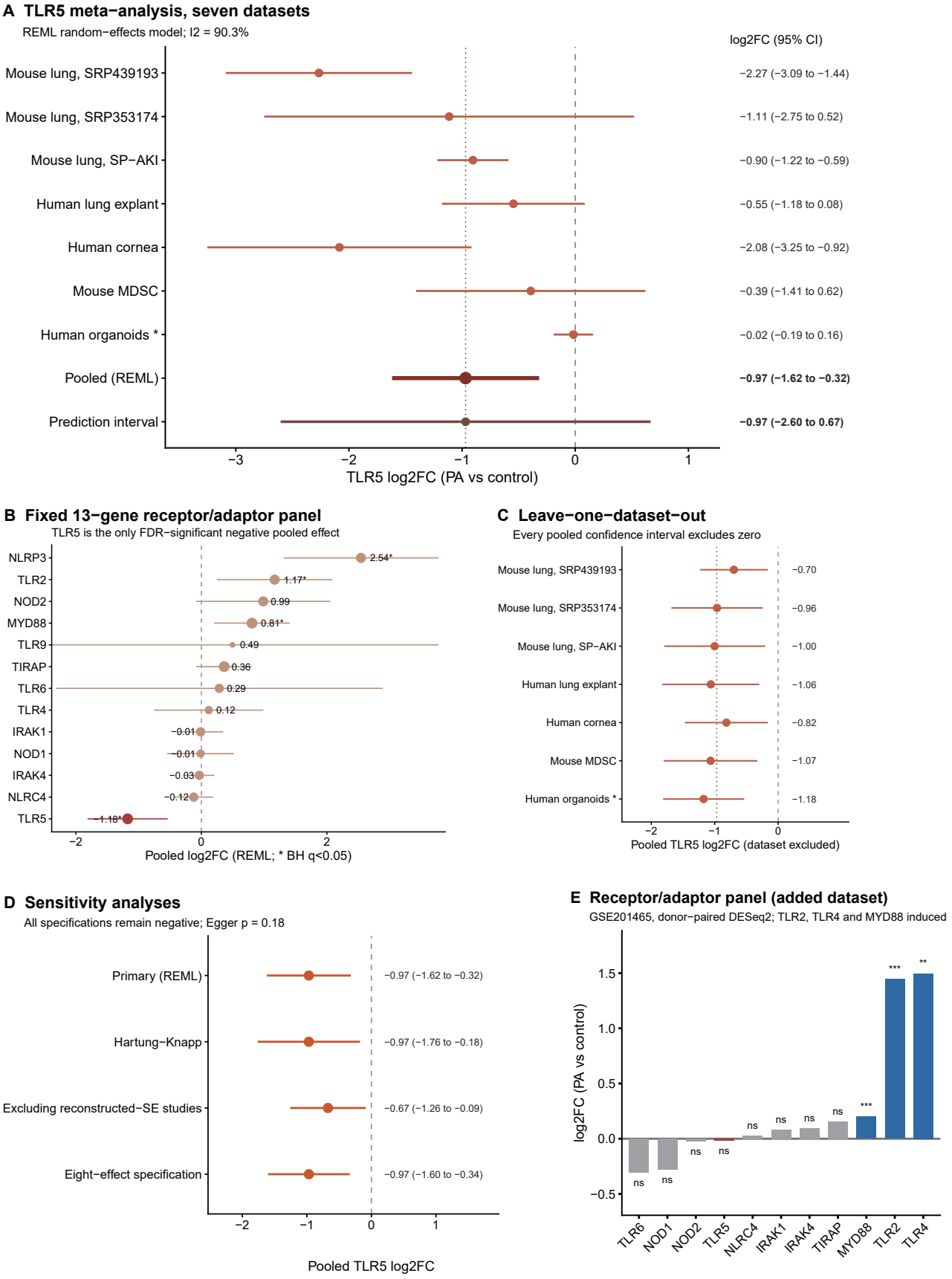

A TLR5 meta−analysis, seven datasets
REML random−effects model; I2 = 90.3%
log2FC (95% CI)
Mouse lung, SRP439193
Mouse lung, SRP353174
Mouse lung, SP−AKI
Human lung explant
Human cornea
Mouse MDSC
Human organoids *
Pooled (REML)
Prediction interval
−2.27 (−3.09 to −1.44)
−1.11 (−2.75 to 0.52)
−0.90 (−1.22 to −0.59)
−0.55 (−1.18 to 0.08)
−2.08 (−3.25 to −0.92)
−0.39 (−1.41 to 0.62)
−0.02 (−0.19 to 0.16)
−0.97 (−1.62 to −0.32)
−0.97 (−2.60 to 0.67)
−3
−2
−1
0
1
TLR5 log2FC (PA vs control)
B Fixed 13−gene receptor/adaptor panel
TLR5 is the only FDR−significant negative pooled effect
NLRP3
TLR2
NOD2
MYD88
TLR9
TIRAP
TLR6
TLR4
IRAK1
NOD1
IRAK4
NLRC4
TLR5
2.54*
1.17*
0.99
0.81*
0.49
0.36
0.29
0.12
−0.01
−0.01
−0.03
−0.12
−1.18*
−2
0
2
Pooled log2FC (REML; * BH q<0.05)
C Leave−one−dataset−out
Every pooled confidence interval excludes zero
Mouse lung, SRP439193
Mouse lung, SRP353174
Mouse lung, SP−AKI
Human lung explant
Human cornea
Mouse MDSC
Human organoids *
−0.70
−0.96
−1.00
−1.06
−0.82
−1.07
−1.18
−2
−1
0
Pooled TLR5 log2FC (dataset excluded)
D Sensitivity analyses
All specifications remain negative; Egger p = 0.18
Primary (REML)
Hartung−Knapp
Excluding reconstructed−SE studies
Eight−effect specification
−0.97 (−1.62 to −0.32)
−0.97 (−1.76 to −0.18)
−0.67 (−1.26 to −0.09)
−0.97 (−1.60 to −0.34)
−2
−1
0
1
Pooled TLR5 log2FC
E Receptor/adaptor panel (added dataset)
GSE201465, donor−paired DESeq2; TLR2, TLR4 and MYD88 induced
log2FC (PA vs control)
1.5
1.0
0.5
0.0
−0.5
ns
ns
ns
ns
ns
ns
ns
ns
***
***
**
TLR6
NOD1
NOD2
TLR5
NLRC4
IRAK1
IRAK4
TIRAP
MYD88
TLR2
TLR4